Command

Season: 3 · Behavioral · Category: GoF · Level: Intermediate

Intent

- Encapsulate a request as an object (queue, undo, log).

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Restaurant ticket the kitchen executes and can void.

When to Use

- See Intent.
- The design force named in Intent shows up more than once.

When Not to Use

- Overkill for one-off behavior.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.behavioral.command;

import java.util.ArrayDeque;
import java.util.Deque;

interface Command { void execute(); void undo(); }

final class Editor {
    private final StringBuilder text = new StringBuilder();
    void append(String s) { text.append(s); }
    void deleteLast(int n) { text.delete(Math.max(0, text.length() - n), text.length()); }
    String text() { return text.toString(); }
}

record WriteCommand(Editor editor, String chunk) implements Command {
    public void execute() { editor.append(chunk); }
    public void undo() { editor.deleteLast(chunk.length()); }
}

/** Invoker with undo stack. */
public final class CommandBus {
    private final Deque<Command> history = new ArrayDeque<>();

    public void run(Command command) {
        command.execute();
        history.push(command);
    }
```

Classic Implementation (2)

```
    public void undo() {
        if (!history.isEmpty()) history.pop().undo();
    }
}
```

Modern Java 25 / Functional

- Java 25 pattern matching often replaces Visitor/Interpreter boilerplate.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/behavioral/command/
```

Pitfalls

- Prefer sealed + switch over classic Visitor in Java 25.
- Strategy as a function value.
- Observer: unsubscribe to avoid leaks.

Interview Q&A

- Q: Key interview angle for Command? A: Name the force (undo, open/closed, open for extension) and a modern alternative.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Encapsulate a request as an object (queue, undo, log).
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Restaurant ticket the kitchen executes and can void. [0:40] Intent — State the design force in one s...